0. Scope & Methodology

This report estimates correctness CONDITIONAL ON NON-TIMEOUT COMPLETION. It should NOT be interpreted as overall execution success rate. Timeout is a major failure mode (v4 Section 7: 47% of prefill failures are TIMEOUT) and is analyzed separately, not deleted.

Timeout definition (matching v4): wall > 600s OR stdout/stderr mentions 'timeout'/'killed'/'timed out'. Dedup: (task, pool, noise, seed), last wins.

First20: sb_exec_20t.jsonl (seed=0) + sb_prefill_n5.jsonl (seeds 1-4). 6 never-executed tasks excluded (court-form-filling, crystallographic-wyckoff, data-to-d3, dynamic-object-aware-egomotion, edit-pdf, exoplanet-detection-period). 14 evaluable tasks remain. Pool sizes: [5, 10, 20, 50].

Smoke10/20: seeds [0-4], pool sizes [5, 10, 20, 50, 100]. Smoke20: 30 design skips at pool=5 for |GT|=6 tasks excluded.

1. Data Reconciliation

First20 Reconciliation

Item	Expected	Actual	Note
Total tasks	20	20	Original set
Never-executed (wall=None)	6	6	Heavy Dockerfile / host-unsafe
Evaluable tasks	14	14	Matches v4 Section 3
Configs per task	60	60	4 pools x 3 noise x 5 seeds
Total configs (N=5)	840	840	14 x 60

Smoke10 Reconciliation

Item	Expected	Actual	Note
Total tasks	10	10	
Expected per task	75	75	5 pools x 3 noise x 5 seeds
Incomplete tasks	0	2	jpg-ocr-stat(52); glm-lake-mendota(68)

Smoke20 Reconciliation

Item	Expected	Actual	Note
Total tasks	20	20	
Expected per task	75	75	5 pools x 3 noise x 5 seeds
Design skips (|GT|>pool)	0	30	pool=5 for python-scala-translation, travel-planning
Valid rows after skip	1500	1470	

2. Waterfall: Raw to Conditional Rate

Decomposing the path from the 50-task report's raw row-weighted rate to this report's conditional non-timeout rate. Each step shows what changed.

First20

Step	Denominator	Pass	Rate
Raw (all deduped configs)	840	503	59.9%
Timeout failures removed	-123		
Conditional non-timeout	717	503	70.2%

Smoke10

Step	Denominator	Pass	Rate
Raw (all deduped configs)	720	300	41.7%
Timeout failures removed	-137		
Conditional non-timeout	583	300	51.5%

Smoke20

Step	Denominator	Pass	Rate
Raw (all deduped configs)	1470	321	21.8%
Timeout failures removed	-554		
Conditional non-timeout	916	321	35.0%

3. Per-Task Four-Metric Diagnostic

Each task shows four metrics. Tasks classified by (completion rate, conditional correctness):

reliable-solved: high pass, high completion (>70% both)
timeout-prone-solved: correct when completed, but often times out
wrong-result-hard: completes but produces wrong results
true-unsolved: fails even when it completes

First20

Task	pass/total	TO/total	pass/non-TO	compl%	Tier
3d-scan-calc	59/60 (98%)	0/60 (0%)	59/60 (98%)	100%	reliable-solved
dialogue-parser	58/60 (97%)	0/60 (0%)	58/60 (97%)	100%	reliable-solved
dapt-intrusion-detection	57/60 (95%)	0/60 (0%)	57/60 (95%)	100%	reliable-solved
energy-market-pricing	57/60 (95%)	2/60 (3%)	57/58 (98%)	97%	reliable-solved
earthquake-plate-calculation	54/60 (90%)	0/60 (0%)	54/60 (90%)	100%	reliable-solved
citation-check	52/60 (87%)	0/60 (0%)	52/60 (87%)	100%	reliable-solved
econ-detrending-correlation	50/60 (83%)	0/60 (0%)	50/60 (83%)	100%	reliable-solved
earthquake-phase-association	46/60 (77%)	11/60 (18%)	46/49 (94%)	82%	reliable-solved
exceltable-in-ppt	21/60 (35%)	0/60 (0%)	21/60 (35%)	100%	mixed
adaptive-cruise-control	20/60 (33%)	38/60 (63%)	20/22 (91%)	37%	TO-prone-solved
energy-ac-optimal-power-flow	19/60 (32%)	8/60 (13%)	19/52 (37%)	87%	mixed
civ6-adjacency-optimizer	6/60 (10%)	38/60 (63%)	6/22 (27%)	37%	mixed
enterprise-information-search	3/60 (5%)	2/60 (3%)	3/58 (5%)	97%	wrong-result
azure-bgp-oscillation-route-leak	1/60 (2%)	24/60 (40%)	1/36 (3%)	60%	wrong-result

Smoke10

Task	pass/total	TO/total	pass/non-TO	compl%	Tier
hvac-control	68/75 (91%)	0/75 (0%)	68/75 (91%)	100%	reliable-solved
software-dependency-audit	58/75 (77%)	3/75 (4%)	58/72 (81%)	96%	reliable-solved
mario-coin-counting	43/75 (57%)	0/75 (0%)	43/75 (57%)	100%	mixed
flood-risk-analysis	38/75 (51%)	7/75 (9%)	38/68 (56%)	91%	mixed
pptx-reference-formatting	38/75 (51%)	4/75 (5%)	38/71 (54%)	95%	mixed
sec-financial-report	32/75 (43%)	0/75 (0%)	32/75 (43%)	100%	mixed
r2r-mpc-control	20/75 (27%)	35/75 (47%)	20/40 (50%)	53%	TO-prone-solved
jpg-ocr-stat	2/52 (4%)	49/52 (94%)	2/3 (67%)	6%	TO-prone-solved
financial-modeling-qa	1/75 (1%)	2/75 (3%)	1/73 (1%)	97%	wrong-result
glm-lake-mendota	0/68 (0%)	37/68 (54%)	0/31 (0%)	46%	true-unsolved

Smoke20

Task	pass/total	TO/total	pass/non-TO	compl%	Tier
offer-letter-generator	69/75 (92%)	0/75 (0%)	69/75 (92%)	100%	reliable-solved
sales-pivot-analysis	68/75 (91%)	0/75 (0%)	68/75 (91%)	100%	reliable-solved
grid-dispatch-operator	60/75 (80%)	2/75 (3%)	60/73 (82%)	97%	reliable-solved
lake-warming-attribution	43/75 (57%)	0/75 (0%)	43/75 (57%)	100%	mixed
powerlifting-coef-calc	25/75 (33%)	2/75 (3%)	25/73 (34%)	97%	mixed
fix-druid-loophole-cve	23/75 (31%)	51/75 (68%)	23/24 (96%)	32%	TO-prone-solved
travel-planning	16/60 (27%)	4/60 (7%)	16/56 (29%)	93%	wrong-result
protein-expression-analysis	14/75 (19%)	8/75 (11%)	14/67 (21%)	89%	wrong-result
reserves-at-risk-calc	3/75 (4%)	67/75 (89%)	3/8 (38%)	11%	mixed
video-filler-word-remover	0/75 (0%)	17/75 (23%)	0/58 (0%)	77%	wrong-result
weighted-gdp-calc	0/75 (0%)	2/75 (3%)	0/73 (0%)	97%	wrong-result
setup-fuzzing-py	0/75 (0%)	74/75 (99%)	0/1 (0%)	1%	tiny-sample

fix-build-agentops	0/75 (0%)	31/75 (41%)	0/44 (0%)	59%	wrong-result
pg-essay-to-audiobook	0/75 (0%)	51/75 (68%)	0/24 (0%)	32%	true-unsolved
xlsx-recover-data	0/75 (0%)	1/75 (1%)	0/74 (0%)	99%	wrong-result
python-scala-translation	0/60 (0%)	60/60 (100%)	N/A	0%	tiny-sample
shock-analysis-demand	0/75 (0%)	73/75 (97%)	0/2 (0%)	3%	tiny-sample
shock-analysis-supply	0/75 (0%)	75/75 (100%)	N/A	0%	tiny-sample
fix-build-google-auto	0/75 (0%)	35/75 (47%)	0/40 (0%)	53%	wrong-result
simpo-code-reproduction	0/75 (0%)	1/75 (1%)	0/74 (0%)	99%	wrong-result

4. Pool x Noise Grid (Raw AND Conditional)

Each cell shows TWO rates: raw pass/total | conditional pass/non-timeout. This prevents the conditional rate from hiding timeout-dominated cells.

First20 (raw | conditional)

Pool\Noise	easy	hard	random	ALL
5	40/70(57%) | 40/58(68%)	41/70(58%) | 41/60(68%)	43/70(61%) | 43/64(67%)	124/210(59%) | 124/182(68%)
10	36/70(51%) | 36/60(60%)	41/70(58%) | 41/61(67%)	40/70(57%) | 40/58(68%)	117/210(55%) | 117/179(65%)
20	42/70(60%) | 42/60(70%)	42/70(60%) | 42/58(72%)	42/70(60%) | 42/59(71%)	126/210(60%) | 126/177(71%)
50	41/70(58%) | 41/61(67%)	48/70(68%) | 48/62(77%)	47/70(67%) | 47/56(83%)	136/210(64%) | 136/179(75%)
100	--	--	--	--
ALL	159/280(56%) | 159/239(66%)	172/280(61%) | 172/241(71%)	172/280(61%) | 172/237(72%)	503/840(59%) | 503/717(70%)

Smoke10 (raw | conditional)

Pool\Noise	easy	hard	random	ALL
5	20/50(40%) | 20/42(47%)	23/50(46%) | 23/37(62%)	21/50(42%) | 21/37(56%)	64/150(42%) | 64/116(55%)
10	14/45(31%) | 14/36(38%)	20/45(44%) | 20/35(57%)	23/49(46%) | 23/39(58%)	57/139(41%) | 57/110(51%)
20	18/48(37%) | 18/39(46%)	18/45(40%) | 18/39(46%)	19/45(42%) | 19/43(44%)	55/138(39%) | 55/121(45%)
50	24/50(48%) | 24/39(61%)	17/50(34%) | 17/41(41%)	19/50(38%) | 19/39(48%)	60/150(40%) | 60/119(50%)
100	22/45(48%) | 22/38(57%)	21/48(43%) | 21/42(50%)	21/50(42%) | 21/37(56%)	64/143(44%) | 64/117(54%)
ALL	98/238(41%) | 98/194(50%)	99/238(41%) | 99/194(51%)	103/244(42%) | 103/195(52%)	300/720(41%) | 300/583(51%)

Smoke20 (raw | conditional)

Pool\Noise	easy	hard	random	ALL
5	28/90(31%) | 28/55(50%)	22/90(24%) | 22/54(40%)	17/90(18%) | 17/59(28%)	67/270(24%) | 67/168(39%)
10	24/100(24%) | 24/61(39%)	22/100(22%) | 22/57(38%)	20/100(20%) | 20/58(34%)	66/300(22%) | 66/176(37%)
20	24/100(24%) | 24/63(38%)	19/100(19%) | 19/57(33%)	20/100(20%) | 20/66(30%)	63/300(21%) | 63/186(33%)
50	25/100(25%) | 25/64(39%)	18/100(18%) | 18/62(29%)	23/100(23%) | 23/63(36%)	66/300(22%) | 66/189(34%)
100	20/100(20%) | 20/69(28%)	18/100(18%) | 18/67(26%)	21/100(21%) | 21/61(34%)	59/300(19%) | 59/197(29%)
ALL	121/490(24%) | 121/312(38%)	99/490(20%) | 99/297(33%)	101/490(20%) | 101/307(32%)	321/1470(21%) | 321/916(35%)

5. V4 Consistency Check (First20, seed=0)

Reproducing v4 Section 8 using seed=0 only, 14 evaluable tasks, dedup by (task, pool, noise), NO timeout exclusion.

Tasks in v4 check: 14 (3d-scan-calc, adaptive-cruise-control, azure-bgp-oscillation-route-leak, citation-check, civ6-adjacency-optimizer...)

Total deduped configs: 168

Noise Mode (v4 ref: random=58.9%, hard=60.7%, easy=57.1%)

Noise	N	Pass	Rate	v4 Rate	Match
easy	56	32	57.1%	57.1%	exact
hard	56	34	60.7%	60.7%	exact
random	56	33	58.9%	58.9%	exact

Pool Size (v4 ref: 5=59.5%, 10=59.5%, 20=61.9%, 50=54.8%)

Pool	N	Pass	Rate	v4 Rate	Match
5	42	25	59.5%	59.5%	exact
10	42	25	59.5%	59.5%	exact
20	42	26	61.9%	61.9%	exact
50	42	23	54.8%	54.8%	exact

6. Cross-Batch Summary

By Noise Mode

Batch	easy raw | cond	hard raw | cond	random raw | cond
First20	56.8% | 66.5%	61.4% | 71.4%	61.4% | 72.6%
Smoke10	41.2% | 50.5%	41.6% | 51.0%	42.2% | 52.8%
Smoke20	24.7% | 38.8%	20.2% | 33.3%	20.6% | 32.9%

By Pool Size

Batch	5 raw | cond	10 raw | cond	20 raw | cond	50 raw | cond	100 raw | cond
First20	59.0% | 68.1%	55.7% | 65.4%	60.0% | 71.2%	64.8% | 76.0%	--
Smoke10	42.7% | 55.2%	41.0% | 51.8%	39.9% | 45.5%	40.0% | 50.4%	44.8% | 54.7%
Smoke20	24.8% | 39.9%	22.0% | 37.5%	21.0% | 33.9%	22.0% | 34.9%	19.7% | 29.9%

7. Conclusion

This report decomposes execution failures into 'wrong answer' vs 'timeout'. It does NOT evaluate whether skills improve performance -- that requires Noskill/GT-only/Noise-only controls (v4 Section 11).

After dedup and exclusion of 814 timeout failures and 30 design skips, conditional non-timeout correctness is 1124/2216 = 50.7%. End-to-end execution success is 1124/3030 = 37.1%.

We do not observe a clear monotonic aggregate trend by pool size or noise mode in either the raw or conditional tables. However, formal inference requires task- and seed-controlled analysis (logistic mixed model). We cannot claim 'no significant effect' without running statistical tests.

Timeout frequency varies strongly by task. Several tasks appear highly accurate conditional on completion while failing end-to-end due to frequent timeouts (see Section 3 'TO-prone-solved' tier). These should NOT be interpreted as 'easy tasks' -- timeout is their dominant failure mode.

Per v4 Section 9: the paired test across 14 First20 tasks is non-significant (p=0.14, CI spans zero). The aggregate 63% vs 41% advantage dissolves when properly accounting for pseudoreplication and task-level pairing.